

Geometry Data Operations and Troubleshooting

HES 505 Fall 2025: Session 11

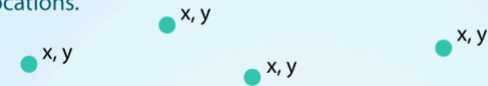
Matt Williamson

The Vector Data Model

- Coordinates define the **Vertices** (i.e., discrete x-y locations) that comprise the geometry
- The organization of those vertices define the *shape* of the vector
- General types: points, lines, polygons

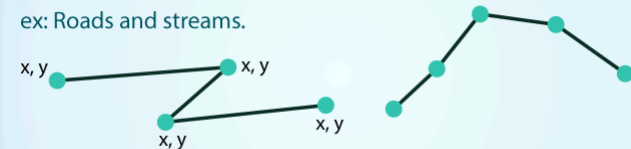
POINTS: Individual **x, y** locations.

ex: Center point of plot locations, tower locations, sampling locations.



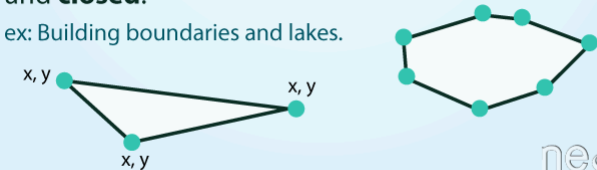
LINES: Composed of many (at least 2) vertices, or points, that are connected.

ex: Roads and streams.



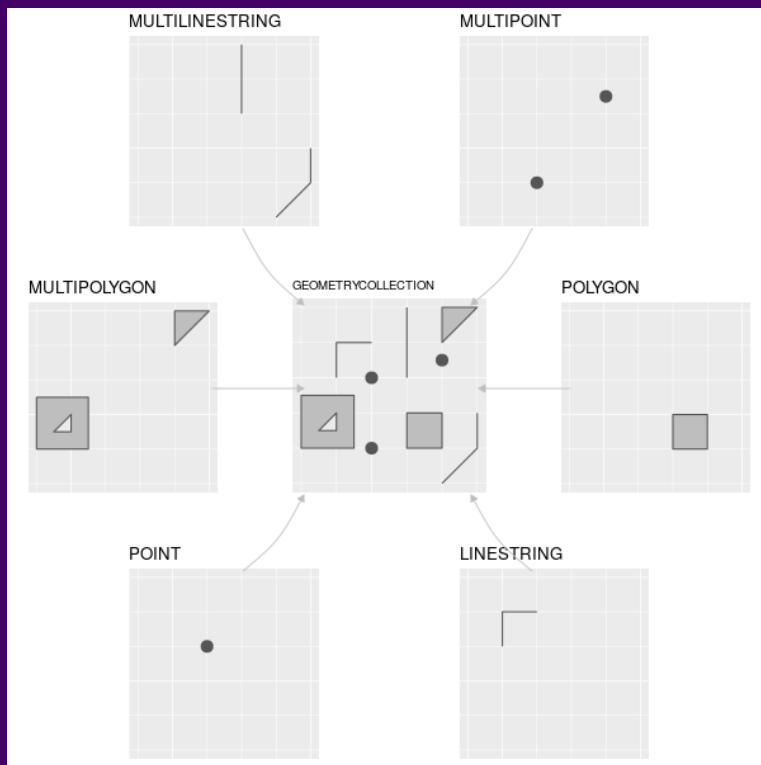
POLYGONS: 3 or more vertices that are connected and **closed**.

ex: Building boundaries and lakes.



neon

Representing vector data in R



- `sf` hierarchy reflects increasing complexity of geometry
 - `st_point`, `st_linestring`, `st_polygon` for single features
 - `st_multi*` for multiple features of the same type
 - `st_geometrycollection` for multiple feature types
 - `st_as_sfc` creates the geometry list column for many `sf` operations

From Lovelace et al.

Revisiting Simple Features

- The **sf** package relies on a simple feature data model to represent geometries
 - hierarchical
 - standardized methods
 - complementary binary and human-readable encoding

type	description
POINT	single point geometry
MULTIPOINT	set of points
LINESTRING	single linestring (two or more points connected by straight lines)
MULTILINESTRING	set of linestrings
POLYGON	exterior ring with zero or more inner rings, denoting holes
MULTIPOLYGON	set of polygons
GEOMETRYCOLLECTION	set of the geometries above

Standardized Methods

We can categorize **sf** operations based on what they return and / or how many geometries they accept as input.

- *Output Categories*

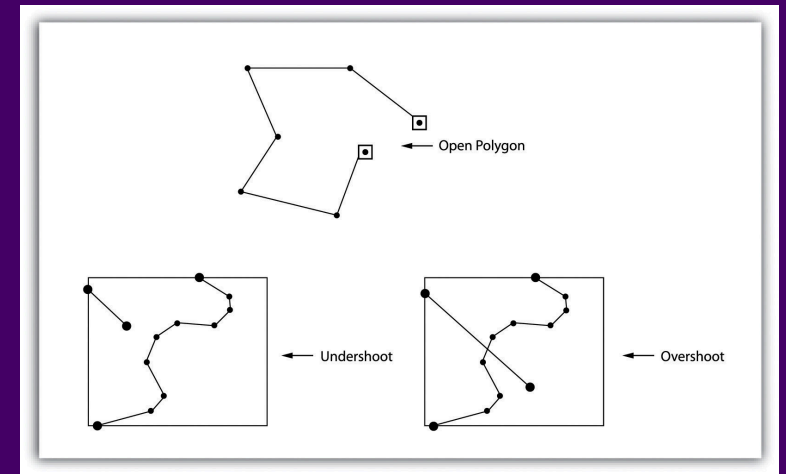
- **Predicates:** evaluate a logical statement asserting that a property is **TRUE**
- **Measures:** return a numeric value with units based on the units of the CRS
- **Transformations:** create new geometries based on input geometries.

- *Input Geometries*

- **Unary:** operate on a single geometry at a time (meaning that if you have a **MULTI*** object the function works on each geometry individually)
- **Binary:** operate on pairs of geometries
- **n-ary:** operate on sets of geometries

Common Problems with Vector Data

- Vectors and scale
- Slivers and overlaps
- Undershoots and overshoots
- Self-intersections and rings



Topology Errors - Saylor Acad.

We'll use `st_is_valid()` (a *predicate*) to check this, but fixing can be tricky

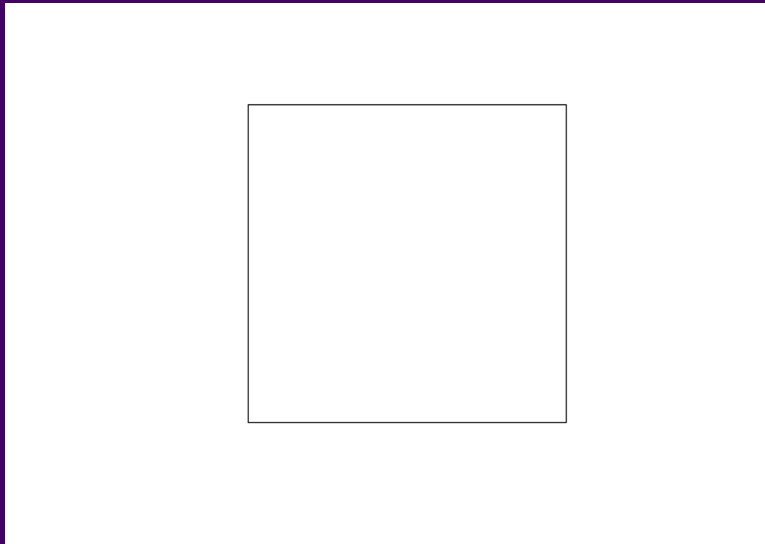
Fixing Problematic Topology

- `st_make_valid()` (a *transformer*) for simple cases
- `st_buffer` (also a *transformer*) with `dist=0`

Fixing geometries

- When `all(st_is_valid(your.shapefile))` returns `FALSE`

Fixing geometries with `st_make_valid`



```
1  ```{r}  
2  y <- x %>% st_make_valid()  
3  st_is_valid(y)  
4  ```
```

```
[1] TRUE
```

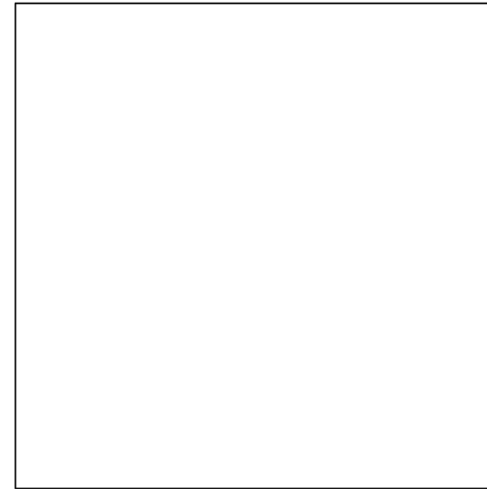
Fixing Geometries with `st_buffer`

-**st_buffer** enforces valid geometries as an output

- Setting a 0 distance buffer leaves most geometries unchanged
- Not all transformations do this

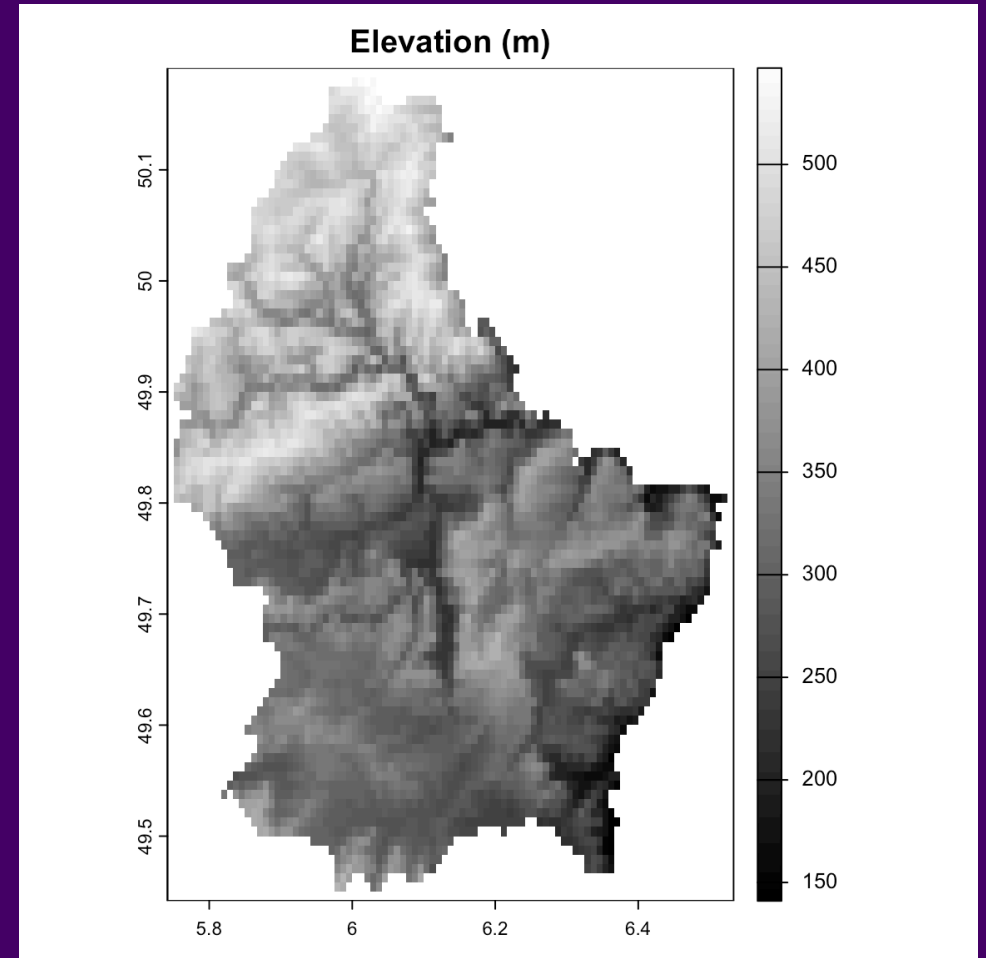
```
1  ```{r}  
2  z <- x %>% st_buffer(., dist=0)  
3  
4  st_is_valid(z)  
5  ```
```

```
[1] TRUE
```



Revisiting the Raster Data Model

- **Vector data** describe the “exact” locations of features on a landscape (including a Cartesian landscape)
- **Raster data** represent spatially continuous phenomena (**NA** is possible)
- Depict the alignment of data on a regular lattice (often a square)
 - Operations mimic those for **matrix** objects in **R**
- Geometry is implicit; the spatial extent and number of rows and columns define the cell size



Rasters with terra

- syntax is different for **terra** compared to **sf**
- Representation in **Environment** is also different
- Can break pipes, **Be Explicit**

Rasters by Construction

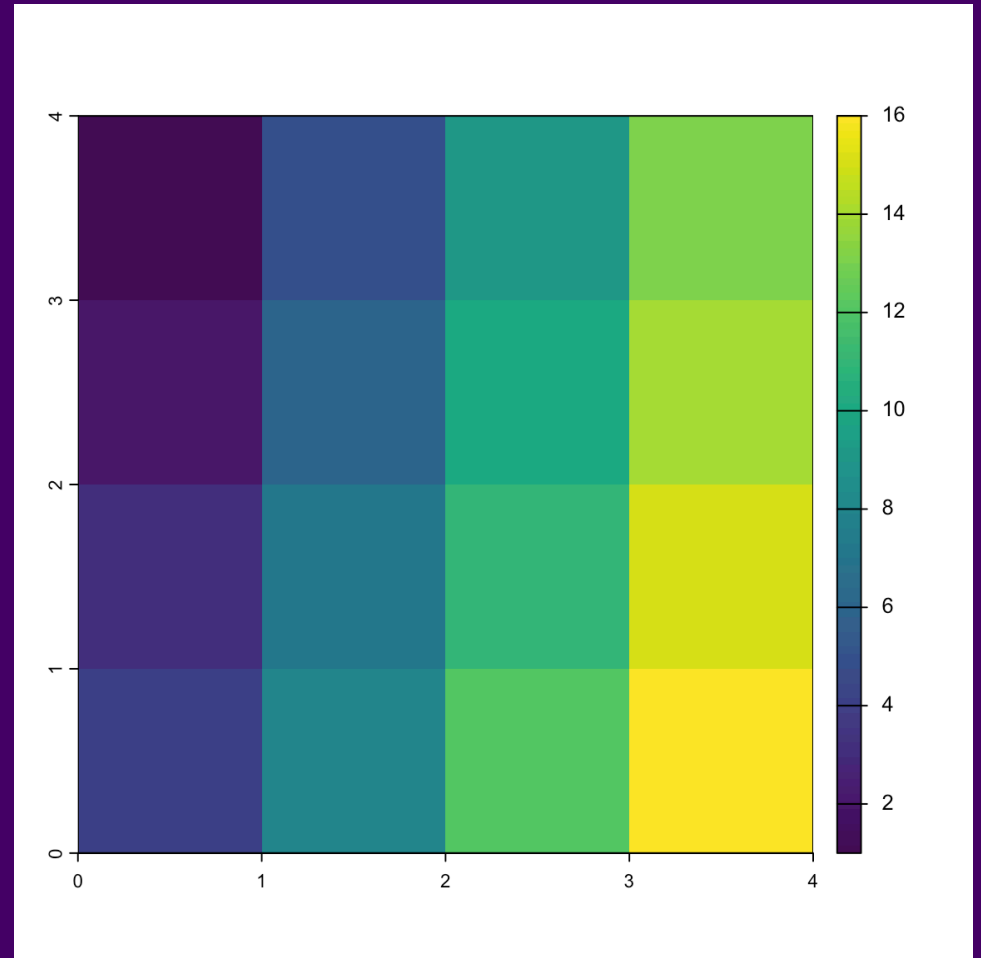
Rasters by Construction

```
1 mtx <- matrix(1:16, nrow=4)
2 mtx
```

```
      [,1] [,2] [,3] [,4]
[1,]     1     5     9    13
[2,]     2     6    10    14
[3,]     3     7    11    15
[4,]     4     8    12    16
```

```
1 rstr <- terra::rast(mtx)
2 rstr
```

```
class      : SpatRaster
size       : 4, 4, 1 (nrow, ncol,
nlyr)
resolution : 1, 1 (x, y)
extent     : 0, 4, 0, 4 (xmin,
xmax, ymin, ymax)
coord. ref.:
source(s)  : memory
name       : lyr.1
min value  :      1
max value  :     16
```



Rasters by Construction: Origin

• **Origin** is the **starting point** of the raster construction process.

• It is the **reference point** for the **coordinates** of the raster cells.

• The **origin** is typically the **top-left corner** of the raster.

• The **origin** is the **starting point** for the **row and column indices**.

• The **origin** is the **starting point** for the **row and column indices**.

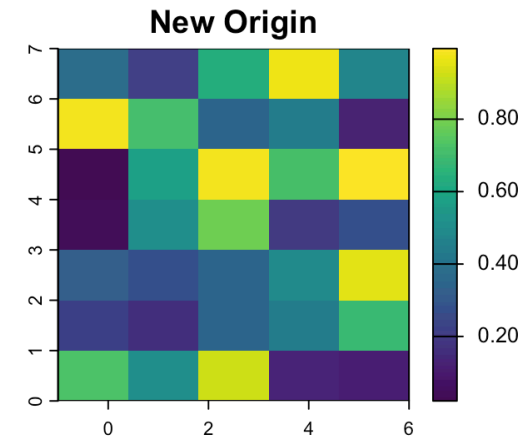
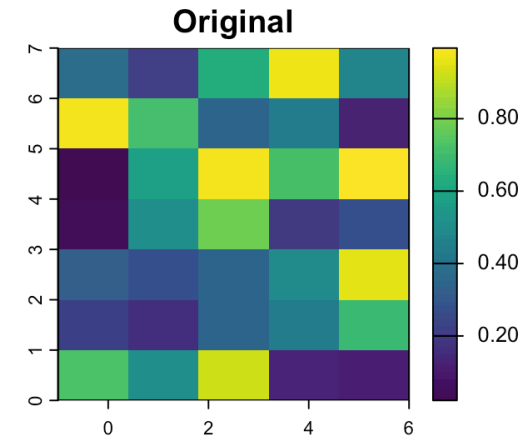
• The **origin** is the **starting point** for the **row and column indices**.

- Origin defines the location of the intersection of the x and y axes

```
1 r <- rast(xmin=-4, xmax = 9.5, ncc  
2 r[] <- runif(ncell(r))  
3 origin(r)
```

```
[1] 0.05 0.00
```

```
1 r2 <- r  
2 origin(r2) <- c(2,2)
```



Rasters by Construction: Resolution

- Geometry is implicit; the spatial extent and number of rows and columns define the cell size
- **Resolution** (**res**) defines the length and width of an individual pixel

```
1 r <- rast(xmin=-4, xmax = 9.5,  
2          ncols=10)  
3 res(r)
```

```
[1] 1.35 1.00
```

```
1 r2 <- rast(xmin=-4, xmax = 5,  
2          ncols=10)  
3 res(r2)
```

```
[1] 0.9 1.0
```

```
1 r <- rast(xmin=-4, xmax = 9.5,  
2          res=c(0.5,0.5))  
3 ncol(r)
```

```
[1] 27
```

```
1 r2 <- rast(xmin=-4, xmax = 9.5,  
2          res=c(5,5))  
3 ncol(r2)
```

```
[1] 3
```

Predicates and measures in terra

Extending predicates

- **Predicates:** evaluate a logical statement asserting that a property is **TRUE**
- **terra** does not follow the same hierarchy as **sf** so a little trickier

Unary predicates in terra

- Can tell us qualities of a raster dataset
- Many similar operations for **SpatVector** class (note use of **.**)

predicate	asks...
is.lonlat	Does the object have a longitude/latitude CRS?
inMemory	is the object stored in memory?
is.factor	Are there categorical layers?
hasValues	Do the cells have values?

Unary predicates in terra

- **global**: tests if the raster covers all longitudes (from -180 to 180 degrees) such that the extreme columns are in fact adjacent
- **perhaps**: If TRUE and the crs is unknown, the method returns TRUE if the coordinates are plausible for longitude/latitude

```
1 r <- rast()  
2 is.lonlat(r)
```

```
[1] TRUE
```

```
1 is.lonlat(r, global=TRUE)
```

```
[1] TRUE
```

```
1 crs(r) <- ""  
2 is.lonlat(r)
```

```
[1] NA
```

```
1 is.lonlat(r, perhaps=TRUE, warn=FALSE)
```

```
[1] TRUE
```

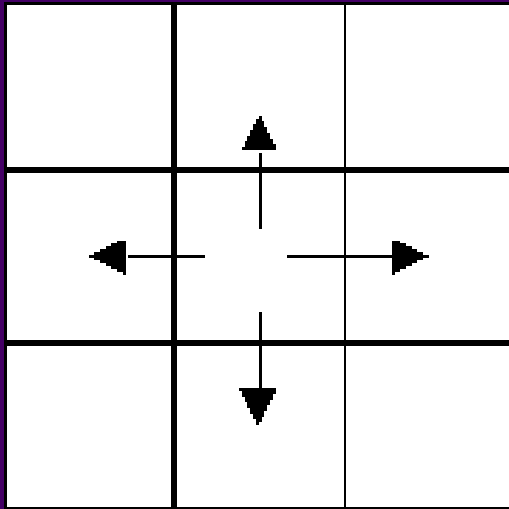
```
1 crs(r) <- "+proj=lcc +lat_1=48 +lat_2=33 +lon_0=-1  
2 is.lonlat(r)
```

```
[1] FALSE
```

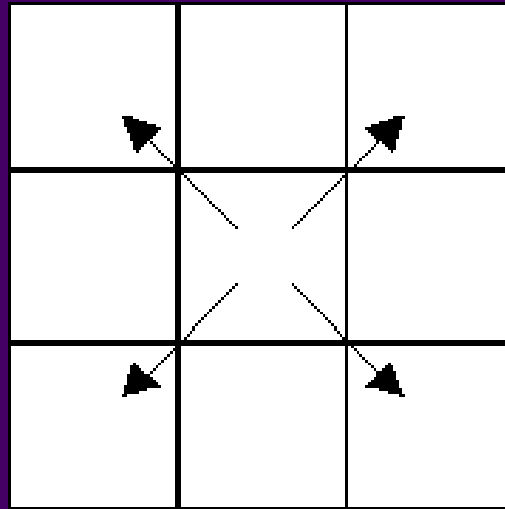
Binary predicates in terra

- Take exactly 2 inputs, return 1 matrix of cell locs where value is **TRUE**
- **adjacent**: identifies cells adjacent to a set of raster cells

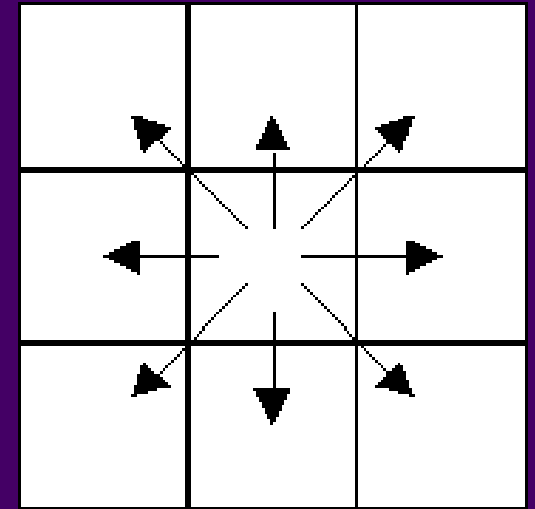
Rooks Case



Bishops Case



Queen's (Kings) Case



Unary measures in terra

- Slightly more flexible than **sf**
- One result for each layer in a stack

measure	returns
cellSize	area of individual cells
expanse	summed area of all cells
values	returns all cell values
ncol	number of columns
nrow	number of rows
ncell	number of cells
res	resolution
ext	minimum and maximum of x and y coords
origin	the origin of a SpatRaster
crs	the coordinate reference system
cats	categories of a categorical raster

Binary measures in terra

- Returns a matrix or **SpatRaster** describing the measure

measure	returns
distance	shortest distance to non-NA or vector object
gridDistance	shortest distance through adjacent grid cells
costDistance	Shortest distance considering cell-varying friction
direction	azimuth to cells that are not NA

